

Fig_rates

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2026-06-10

Functions

Figures

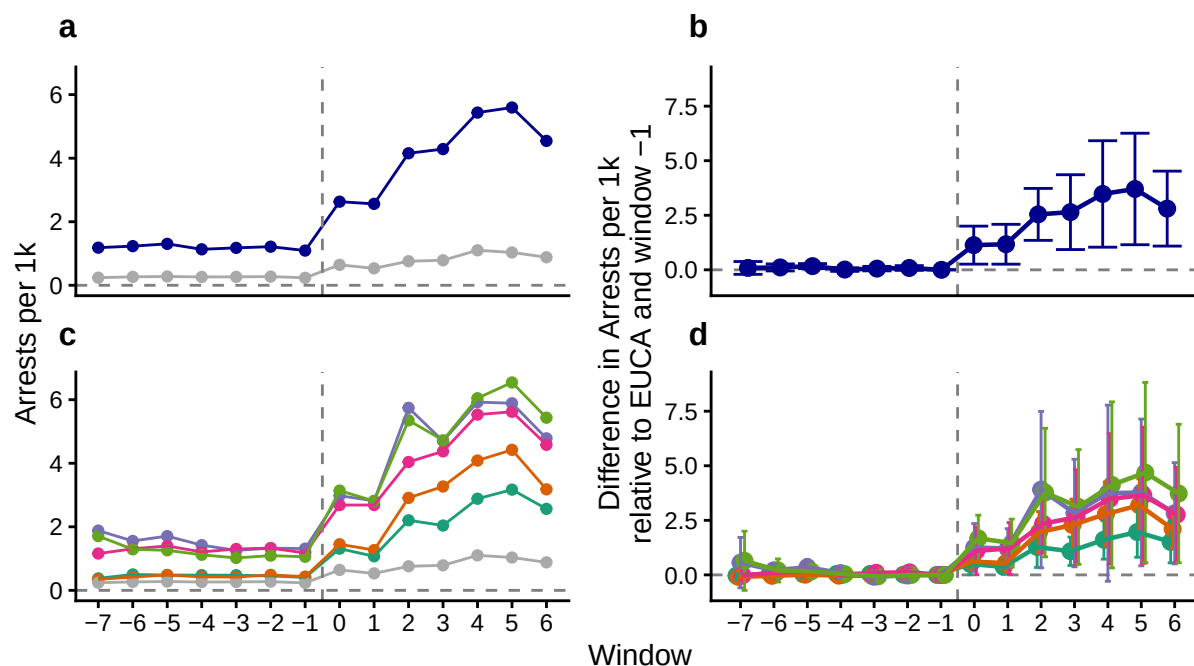
```
## Rows: 231336 Columns: 10
## -- Column specification -----
## Delimiter: ","
## chr (4): state, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.

## # A tibble: 70 x 6
##   window     est      se   lower upper region
##   <dbl>   <dbl> <dbl>   <dbl> <dbl> <chr>
## 1     -7 -0.0336 0.0641 -0.214  0.147 Africa
## 2     -6  0.0639 0.0496 -0.0759  0.204 Africa
## 3     -5  0.0332 0.0698 -0.164  0.230 Africa
## 4     -4  0.0455 0.0509 -0.0979  0.189 Africa
## 5     -3  0.0396 0.0398 -0.0725  0.152 Africa
## 6     -2  0.00992 0.0438 -0.114  0.133 Africa
## 7     -1  0      0      0      0 Africa
## 8      0  0.498  0.173  0.00977  0.986 Africa
## 9      1  0.368  0.115  0.0424  0.694 Africa
## 10     2  1.28   0.332  0.340  2.21 Africa
## # i 60 more rows
## Raw coef names:
##   window::-7:region::nonEUCA window::-6:region::nonEUCA window::-5:region::nonEUCA window::-4:region:
## Wald test, H0: joint nullity of window::-7:region::nonEUCA, window::-6:region::nonEUCA, window::-5:r
##   stat = 3.15277, p-value = 0.004485, on 6 and 1,400 DoF, VCOV: Custom.[1] "Wald Pre-Trend Test"
## $stat
## [1] 3.152774
##
## $p
## [1] 0.004484512
##
## $df1
## [1] 6
##
## $df2
```

```
## [1] 1400
##
## $vcov
## [1] "Custom"
##
## [1] "M bar Results"
## # A tibble: 5 x 5
##       lb      ub method Delta      Mbar
##   <dbl> <dbl> <chr>  <chr>  <dbl>
## 1  0.858  4.32 C-LF   DeltaRM 1
## 2  0.157  5.23 C-LF   DeltaRM 2.2
## 3  0.0846 5.33 C-LF   DeltaRM 2.3
## 4  0.0121 5.40 C-LF   DeltaRM 2.4
## 5 -0.0604 5.50 C-LF   DeltaRM 2.5
## # A tibble: 14 x 6
##   window  est      se  lower  upper region
##   <dbl>  <dbl>  <dbl>  <dbl>  <dbl> <chr>
## 1    -7 0.0870 0.111  -0.206  0.380 nonEUCA
## 2    -6 0.109 0.0608 -0.0518 0.271 nonEUCA
## 3    -5 0.170 0.0436  0.0540 0.285 nonEUCA
## 4    -4 0.0123 0.0309 -0.0697 0.0942 nonEUCA
## 5    -3 0.0553 0.0386 -0.0472 0.158 nonEUCA
## 6    -2 0.0842 0.0406 -0.0235 0.192 nonEUCA
## 7    -1 0      0      0      0 nonEUCA
## 8     0 1.13  0.328  0.260  2.00 nonEUCA
## 9     1 1.17  0.344  0.260  2.08 nonEUCA
## 10    2 2.54  0.449  1.35   3.73 nonEUCA
## 11    3 2.64  0.646  0.929  4.36 nonEUCA
## 12    4 3.48  0.919  1.04   5.91 nonEUCA
## 13    5 3.70  0.963  1.15   6.26 nonEUCA
## 14    6 2.80  0.648  1.09   4.52 nonEUCA

## Warning: Subsetting quosures with `[` is deprecated as of rlang 0.4.0
## Please use `quo_get_expr()` instead.
## This warning is displayed once every 8 hours.

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```



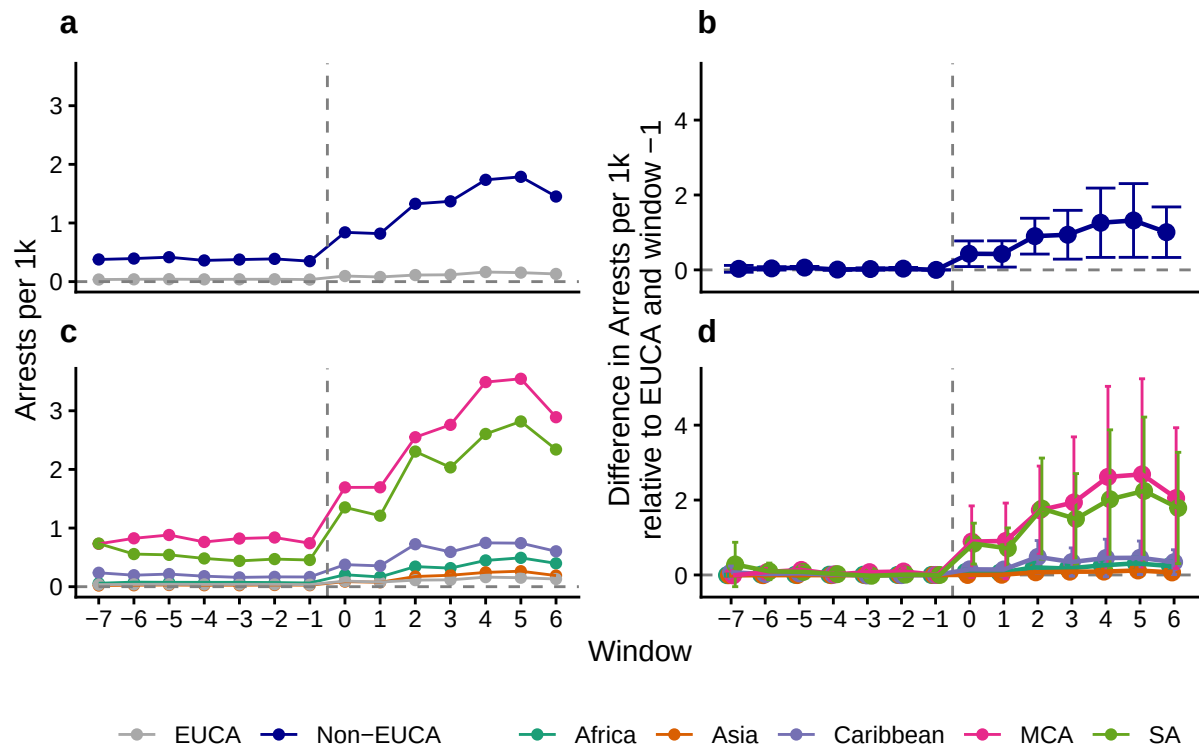
— EUCA — Non-EUCA — Africa — Asia — Caribbean — MCA — SA

```
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by state, region, window, convicted,
##   threat_level, method, and alt_method.
## i Output is grouped by state, region, window, convicted, threat_level, and
##   method.
## i Use `summarise(groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(state, region, window, convicted, threat_level,
##   method, alt_method))` for per-operation grouping (`?dplyr::dplyr_by`)
```

```
## # A tibble: 70 x 6
##   window     est      se   lower upper region
##   <dbl>   <dbl>   <dbl>   <dbl> <dbl> <chr>
## 1      -7 -0.00518 0.0100  -0.0334 0.0230 Africa
## 2      -6  0.0101  0.00756 -0.0112 0.0314 Africa
## 3      -5  0.00543 0.0107  -0.0248 0.0356 Africa
## 4      -4  0.00724 0.00773 -0.0145 0.0290 Africa
## 5      -3  0.00636 0.00611 -0.0109 0.0236 Africa
## 6      -2  0.00180 0.00685 -0.0175 0.0211 Africa
## 7      -1  0      0      0      0      Africa
## 8       0  0.0801  0.0259  0.00709 0.153  Africa
## 9       1  0.0592  0.0180  0.00835 0.110  Africa
## 10      2  0.202   0.0496  0.0620 0.342  Africa
## # i 60 more rows
## # A tibble: 14 x 6
##   window     est      se   lower upper region
##   <dbl>   <dbl>   <dbl>   <dbl> <dbl> <chr>
## 1      -7  0.0286  0.0349  -0.0624 0.120  nonEUCA
## 2      -6  0.0402  0.0174  -0.00510 0.0855 nonEUCA
## 3      -5  0.0613  0.0119   0.0302 0.0925 nonEUCA
```

```
## 4      -4 0.00857 0.00723 -0.0103 0.0274 nonEUCA
## 5      -3 0.0229 0.0117 -0.00746 0.0533 nonEUCA
## 6      -2 0.0335 0.0124 0.00123 0.0657 nonEUCA
## 7      -1 0      0      0      0 nonEUCA
## 8       0 0.431 0.131 0.0886 0.773 nonEUCA
## 9       1 0.425 0.135 0.0748 0.776 nonEUCA
## 10      2 0.901 0.184 0.422 1.38 nonEUCA
## 11      3 0.939 0.250 0.287 1.59 nonEUCA
## 12      4 1.26 0.355 0.333 2.18 nonEUCA
## 13      5 1.32 0.378 0.335 2.30 nonEUCA
## 14      6 1.01 0.259 0.332 1.68 nonEUCA
```

```
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```



```
## # A tibble: 70 x 6
##   window  est    se lower upper region
##   <dbl> <dbl> <dbl> <dbl> <dbl> <chr>
## 1     -7 0.911 0.204 0.514 1.61 Africa
## 2     -6 1.09 0.118 0.784 1.52 Africa
## 3     -5 1.01 0.183 0.603 1.68 Africa
## 4     -4 1.06 0.144 0.707 1.58 Africa
## 5     -3 1.04 0.108 0.768 1.40 Africa
## 6     -2 0.962 0.146 0.639 1.45 Africa
## 7     -1 1      0      1      1 Africa
## 8      0 1.18 0.180 0.715 1.95 Africa
## 9      1 1.16 0.143 0.780 1.74 Africa
## 10     2 1.69 0.208 0.941 3.02 Africa
## # i 60 more rows
```

```

## Raw coef names:
## window::-7:region::nonEUCA window::-6:region::nonEUCA window::-5:region::nonEUCA window::-4:region:
## Wald test, H0: joint nullity of window::-7:region::nonEUCA, window::-6:region::nonEUCA, window::-5:r
## stat = 0.208309, p-value = 0.974287, on 6 and 1,400 DoF, VCOV: Custom.[1] "Wald Pre-Trend Test"
## $stat
## [1] 0.2083088
##
## $p
## [1] 0.9742866
##
## $df1
## [1] 6
##
## $df2
## [1] 1400
##
## $vcov
## [1] "Custom"

## Warning in .ARP_computeCI(betahat = betahat, sigma = sigma, numPrePeriods =
## numPrePeriods, : CI is open at one of the endpoints; CI length may not be
## accurate

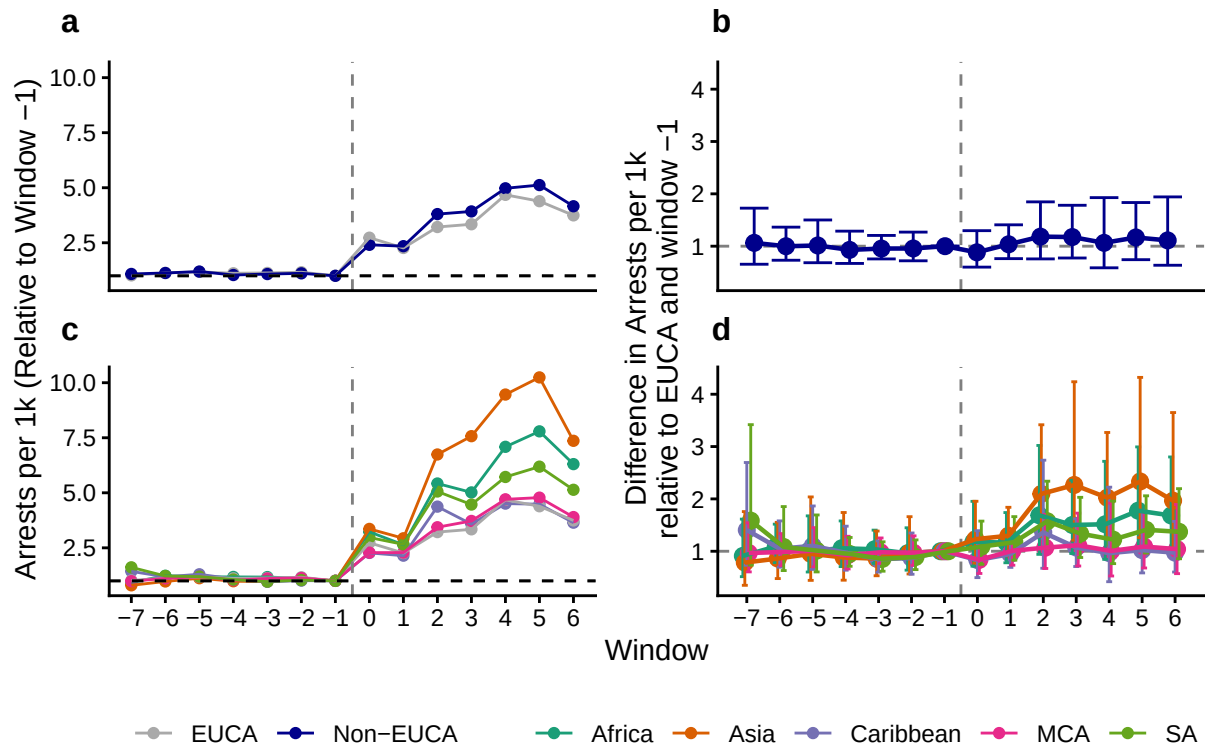
## Warning in .ARP_computeCI(betahat = betahat, sigma = sigma, numPrePeriods =
## numPrePeriods, : CI is open at one of the endpoints; CI length may not be
## accurate

## [1] "M bar Results"
## # A tibble: 5 x 5
##   lb    ub method Delta    Mbar
##   <dbl> <dbl> <chr>   <chr>   <dbl>
## 1 -1.30  1.48 C-LF    DeltaRM  1
## 2 -2.76  2.97 C-LF    DeltaRM  2.2
## 3 -2.88  3.09 C-LF    DeltaRM  2.3
## 4 -3.01  3.14 C-LF    DeltaRM  2.4
## 5 -3.13  3.14 C-LF    DeltaRM  2.5
## # A tibble: 14 x 6
##   window  est      se lower upper region
##   <dbl> <dbl>   <dbl> <dbl> <dbl> <chr>
## 1      -7 1.06  0.180  0.655  1.73 nonEUCA
## 2      -6 0.999 0.116  0.731  1.36 nonEUCA
## 3      -5 1.01  0.146  0.684  1.50 nonEUCA
## 4      -4 0.929 0.121  0.670  1.29 nonEUCA
## 5      -3 0.955 0.0864 0.756  1.21 nonEUCA
## 6      -2 0.956 0.105  0.720  1.27 nonEUCA
## 7      -1 1      0      1      1 nonEUCA
## 8       0 0.882 0.143  0.600  1.30 nonEUCA
## 9       1 1.04  0.114  0.762  1.41 nonEUCA
## 10      2 1.18  0.165  0.757  1.85 nonEUCA
## 11      3 1.17  0.154  0.774  1.78 nonEUCA
## 12      4 1.06  0.220  0.587  1.93 nonEUCA
## 13      5 1.17  0.168  0.742  1.83 nonEUCA
## 14      6 1.11  0.207  0.635  1.94 nonEUCA

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```

```
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## Adding another scale for colour, which will replace the existing scale.
```



Test for significance

```
## Rows: 231336 Columns: 10
## -- Column specification -----
## Delimiter: ","
## chr (4): state, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## [1] "Pre-treatment rates"
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by window and region.
## i Output is grouped by window.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(window, region))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
## # A tibble: 4 x 5
## # Groups:   window [2]
##   window region total_arrests      pop prop
##   <dbl> <chr>      <dbl>    <dbl> <dbl>
## 1     -1 EUCA          208    883000 0.236
## 2     -1 nonEUCA     14015  12829000 1.09
## 3      0 EUCA           724    883000 0.820
## 4      0 nonEUCA    53531.  12829000 4.17
```

```

## [1] 3.544246e-05
## [1] 2.49684
## [1] 1.31343
## [1] 3.68025
Test for significance

## Rows: 231336 Columns: 10
## -- Column specification -----
## Delimiter: ","
## chr (4): state, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.

## [1] "Pre-treatment rates"

## `summarise()` has regrouped the output.
## i Summaries were computed grouped by window and region.
## i Output is grouped by window.
## i Use `summarise(groups = "drop_last")` to silence this message.
## i Use `summarise(by = c(window, region))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.

## # A tibble: 4 x 5
## # Groups:   window [2]
##   window region total_arrests      pop prop
##   <dbl> <chr>         <dbl>    <dbl> <dbl>
## 1     -1 EUCA             208   883000 0.236
## 2     -1 nonEUCA        14015 12829000 1.09
## 3      0 EUCA             724   883000 0.820
## 4      0 nonEUCA       53531. 12829000 4.17

## [1] 0.6477476
## [1] 1.083454
## [1] 0.7682085
## [1] 1.528065

## Rows: 231336 Columns: 10
## -- Column specification -----
## Delimiter: ","
## chr (4): state, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.

## # A tibble: 70 x 6
##   window      est      se lower upper region
##   <dbl>    <dbl> <dbl> <dbl> <dbl> <chr>

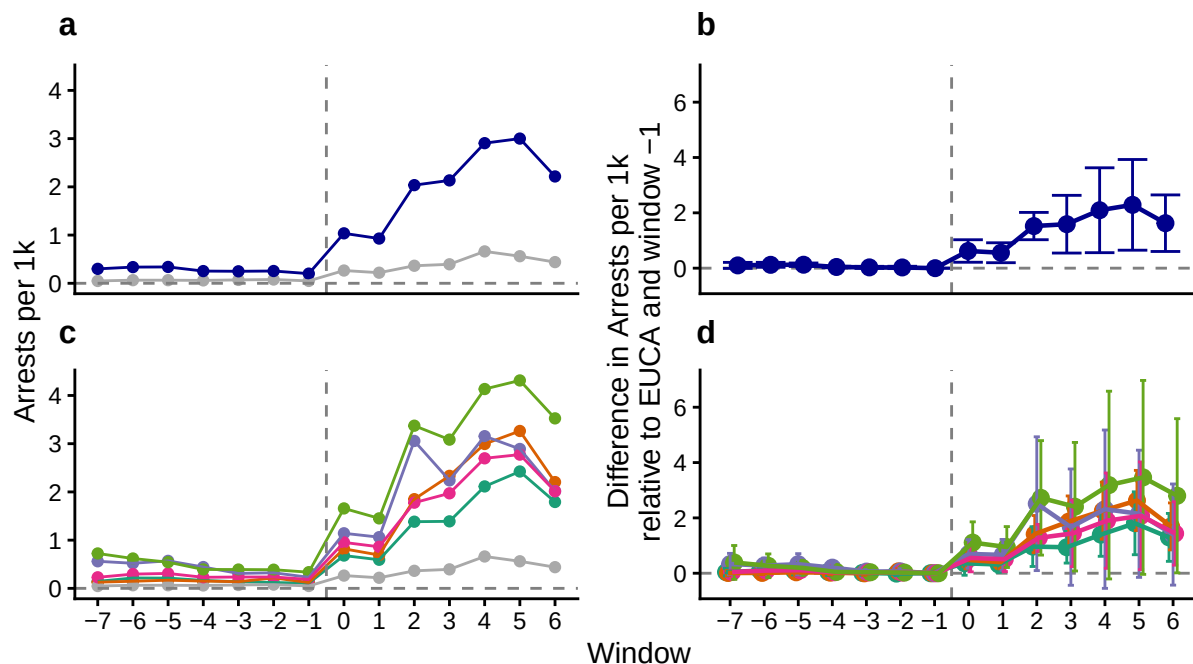
```

```

## 1      -7  0.0399 0.0494 -0.0971 0.177 Africa
## 2      -6  0.0942 0.0433 -0.0258 0.214 Africa
## 3      -5  0.0918 0.0543 -0.0587 0.242 Africa
## 4      -4  0.0479 0.0471 -0.0828 0.179 Africa
## 5      -3  0.0111 0.0304 -0.0732 0.0954 Africa
## 6      -2 -0.0138 0.0270 -0.0888 0.0612 Africa
## 7      -1  0      0      0      0      Africa
## 8       0  0.359  0.155 -0.0721 0.789 Africa
## 9       1  0.317  0.102  0.0337 0.600 Africa
## 10      2  0.962  0.259  0.244  1.68  Africa
## # i 60 more rows
## # A tibble: 14 x 6
##   window    est    se   lower upper region
##   <dbl> <dbl> <dbl>   <dbl> <dbl> <chr>
## 1     -7  0.100  0.0402 -0.00838 0.209 nonEUCA
## 2     -6  0.121  0.0330  0.0313  0.210 nonEUCA
## 3     -5  0.124  0.0231  0.0615  0.187 nonEUCA
## 4     -4  0.0426 0.0103  0.0147  0.0705 nonEUCA
## 5     -3  0.0280 0.0141 -0.0102  0.0662 nonEUCA
## 6     -2  0.0260 0.0155 -0.0158  0.0679 nonEUCA
## 7     -1  0      0      0      0      nonEUCA
## 8      0  0.620  0.151  0.214  1.03  nonEUCA
## 9      1  0.558  0.133  0.198  0.917 nonEUCA
## 10     2  1.52   0.183  1.03   2.01  nonEUCA
## 11     3  1.59   0.386  0.546  2.63  nonEUCA
## 12     4  2.09   0.568  0.560  3.63  nonEUCA
## 13     5  2.29   0.607  0.649  3.93  nonEUCA
## 14     6  1.63   0.378  0.603  2.65  nonEUCA

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

```

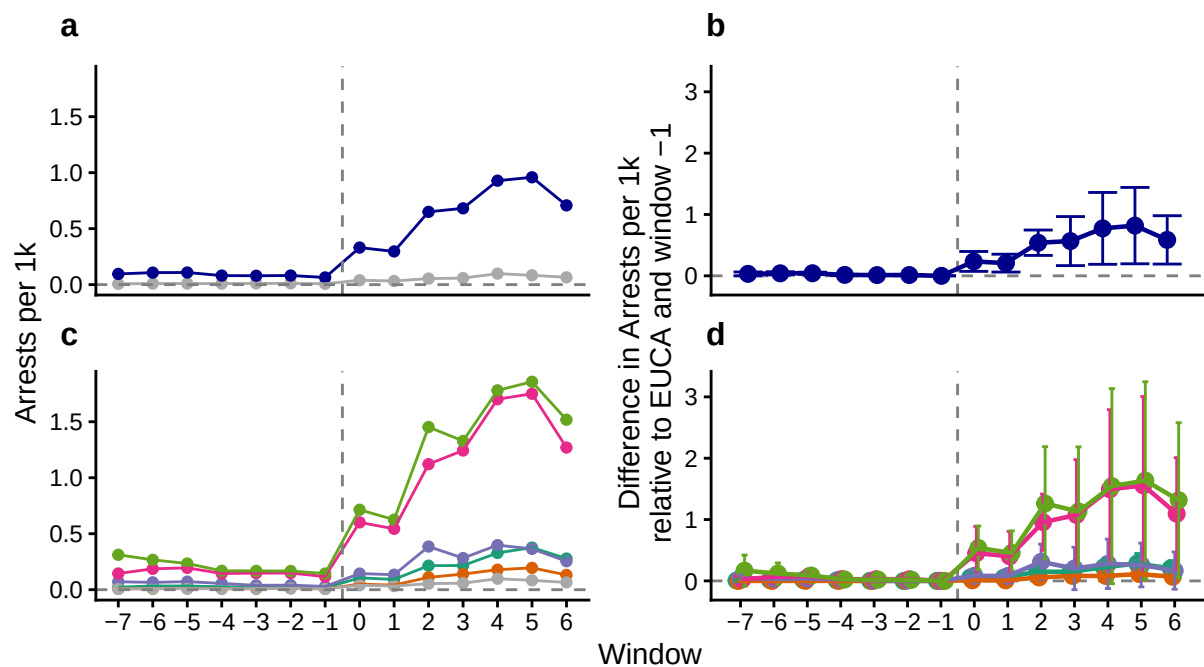
— EUCA — Non-EUCA — Africa — Asia — Caribbean — MCA — SA

```
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by state, region, window, convicted,
##   threat_level, method, and alt_method.
## i Output is grouped by state, region, window, convicted, threat_level, and
##   method.
## i Use `summarise(groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(state, region, window, convicted, threat_level,
##   method, alt_method))` for per-operation grouping (`?dplyr::dplyr_by`)
##   instead.
```

```
## # A tibble: 70 x 6
##   window     est      se    lower  upper region
##   <dbl>   <dbl>   <dbl>   <dbl>   <dbl> <chr>
## 1     -7 0.00618 0.00750 -0.0147 0.0271 Africa
## 2     -6 0.0147 0.00639 -0.00305 0.0325 Africa
## 3     -5 0.0144 0.00819 -0.00846 0.0372 Africa
## 4     -4 0.00751 0.00720 -0.0125 0.0276 Africa
## 5     -3 0.00186 0.00468 -0.0112 0.0149 Africa
## 6     -2 -0.00195 0.00424 -0.0138 0.00986 Africa
## 7     -1 0         0         0         0 Africa
## 8      0 0.0572 0.0236 -0.00859 0.123 Africa
## 9      1 0.0503 0.0155 0.00727 0.0934 Africa
## 10     2 0.152 0.0380 0.0458 0.257 Africa
## # i 60 more rows
## # A tibble: 14 x 6
##   window     est      se    lower  upper region
##   <dbl>   <dbl>   <dbl>   <dbl>   <dbl> <chr>
## 1     -7 0.0318 0.0117 0.00000139 0.0636 nonEUCA
## 2     -6 0.0410 0.00956 0.0150 0.0670 nonEUCA
## 3     -5 0.0421 0.00653 0.0244 0.0599 nonEUCA
```

```
## 4      -4 0.0153 0.00332 0.00632      0.0244 nonEUCA
## 5      -3 0.0124 0.00342 0.00314      0.0217 nonEUCA
## 6      -2 0.0132 0.00373 0.00302      0.0233 nonEUCA
## 7      -1 0      0      0      0      nonEUCA
## 8       0 0.235 0.0596 0.0729      0.397 nonEUCA
## 9       1 0.207 0.0535 0.0617      0.353 nonEUCA
## 10      2 0.540 0.0760 0.333      0.746 nonEUCA
## 11      3 0.566 0.147 0.167      0.966 nonEUCA
## 12      4 0.774 0.215 0.188      1.36 nonEUCA
## 13      5 0.819 0.229 0.196      1.44 nonEUCA
## 14      6 0.586 0.145 0.192      0.980 nonEUCA
```

```
## Scale for colour is already present.
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```

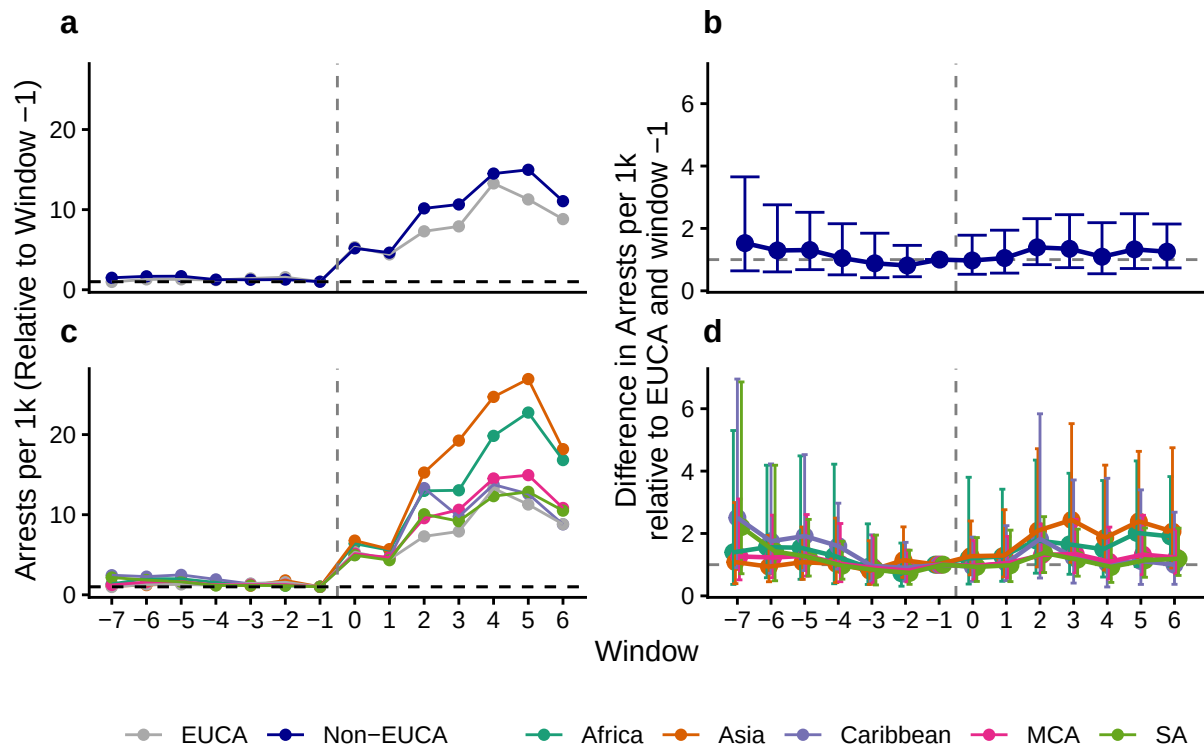


— EUCA — Non-EUCA — Africa — Asia — Caribbean — MCA — SA

```
## # A tibble: 70 x 6
##   window  est    se lower upper region
##   <dbl> <dbl> <dbl> <dbl> <dbl> <chr>
## 1     -7 1.40 0.513 0.367 5.30 Africa
## 2     -6 1.56 0.379 0.582 4.19 Africa
## 3     -5 1.54 0.410 0.531 4.49 Africa
## 4     -4 1.28 0.457 0.390 4.22 Africa
## 5     -3 0.919 0.354 0.366 2.31 Africa
## 6     -2 0.725 0.326 0.310 1.69 Africa
## 7     -1 1      0      1      1 Africa
## 8      0 1.20 0.443 0.379 3.80 Africa
## 9      1 1.26 0.382 0.466 3.42 Africa
## 10     2 1.78 0.344 0.727 4.35 Africa
## # i 60 more rows
```

```
## # A tibble: 14 x 6
##   window  est    se lower upper region
##   <dbl> <dbl> <dbl> <dbl> <dbl> <chr>
## 1     -7 1.53 0.329 0.640 3.65 nonEUCA
## 2     -6 1.29 0.286 0.606 2.76 nonEUCA
## 3     -5 1.31 0.248 0.678 2.52 nonEUCA
## 4     -4 1.05 0.272 0.511 2.15 nonEUCA
## 5     -3 0.881 0.280 0.421 1.85 nonEUCA
## 6     -2 0.811 0.222 0.451 1.46 nonEUCA
## 7     -1 1      0      1      1 nonEUCA
## 8      0 0.972 0.229 0.530 1.78 nonEUCA
## 9      1 1.05 0.233 0.568 1.94 nonEUCA
## 10     2 1.39 0.192 0.838 2.31 nonEUCA
## 11     3 1.35 0.225 0.743 2.44 nonEUCA
## 12     4 1.09 0.262 0.547 2.18 nonEUCA
## 13     5 1.33 0.235 0.714 2.47 nonEUCA
## 14     6 1.25 0.202 0.734 2.14 nonEUCA

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```



```
## Rows: 231336 Columns: 10
## -- Column specification -----
## Delimiter: ","
## chr (4): state, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

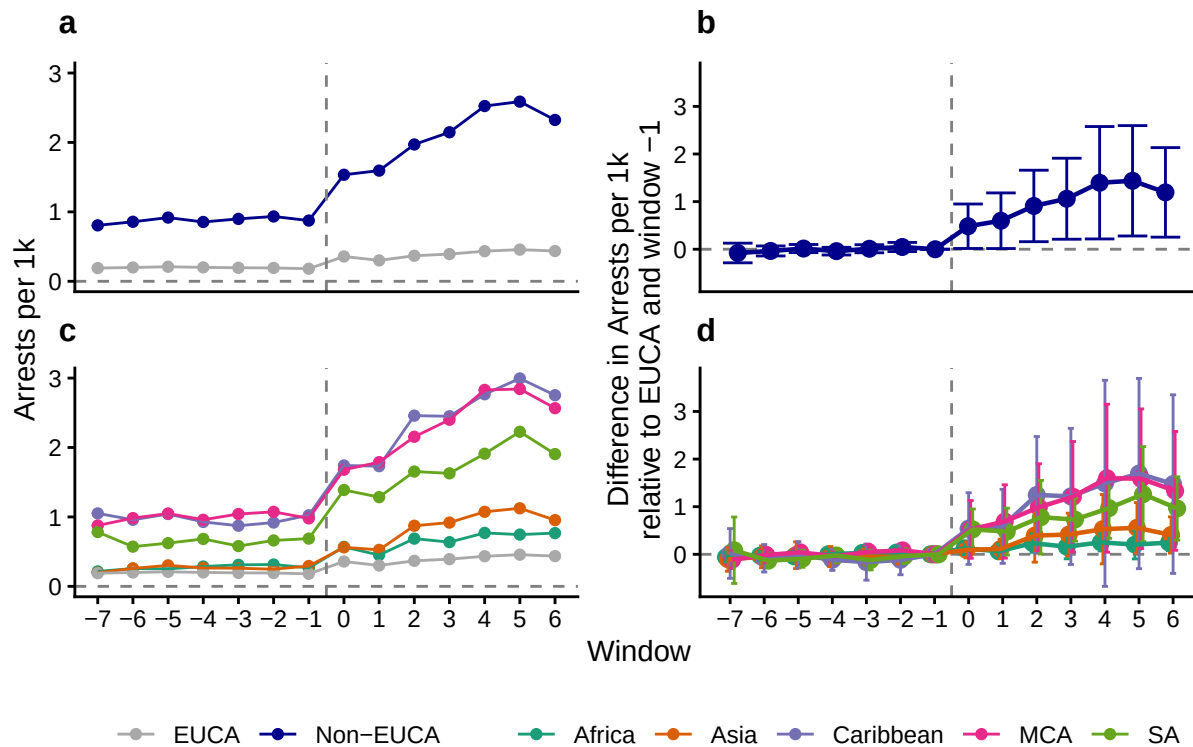
```

## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.

## # A tibble: 70 x 6
##   window      est      se    lower upper region
##   <dbl>    <dbl>  <dbl>    <dbl>  <dbl> <chr>
## 1     -7 -0.0599  0.0359 -0.161  0.0415 Africa
## 2     -6 -0.0278  0.0310 -0.115  0.0599 Africa
## 3     -5 -0.0428  0.0388 -0.152  0.0668 Africa
## 4     -4  0.000118 0.0402 -0.114  0.114  Africa
## 5     -3  0.0300  0.0375 -0.0759  0.136  Africa
## 6     -2  0.0371  0.0483 -0.0993  0.174  Africa
## 7     -1  0      0      0      0      Africa
## 8      0  0.126  0.0463 -0.00474 0.257  Africa
## 9      1  0.0640  0.0539 -0.0881  0.216  Africa
## 10     2  0.234  0.0684  0.0414  0.428  Africa
## # i 60 more rows
## # A tibble: 14 x 6
##   window      est      se    lower upper region
##   <dbl>    <dbl>  <dbl>    <dbl>  <dbl> <chr>
## 1     -7 -0.0784 0.0775 -0.285  0.128  nonEUCA
## 2     -6 -0.0360 0.0394 -0.141  0.0689 nonEUCA
## 3     -5  0.0145 0.0321 -0.0710 0.1000 nonEUCA
## 4     -4 -0.0400 0.0303 -0.121  0.0406 nonEUCA
## 5     -3  0.0103 0.0318 -0.0744 0.0949 nonEUCA
## 6     -2  0.0472 0.0361 -0.0490 0.143  nonEUCA
## 7     -1  0      0      0      0      nonEUCA
## 8      0  0.483  0.176  0.0158 0.950  nonEUCA
## 9      1  0.599  0.220  0.0148 1.18   nonEUCA
## 10     2  0.909  0.282  0.158  1.66   nonEUCA
## 11     3  1.06  0.320  0.210  1.91   nonEUCA
## 12     4  1.40  0.443  0.217  2.58   nonEUCA
## 13     5  1.44  0.436  0.278  2.60   nonEUCA
## 14     6  1.19  0.353  0.254  2.13   nonEUCA

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

```

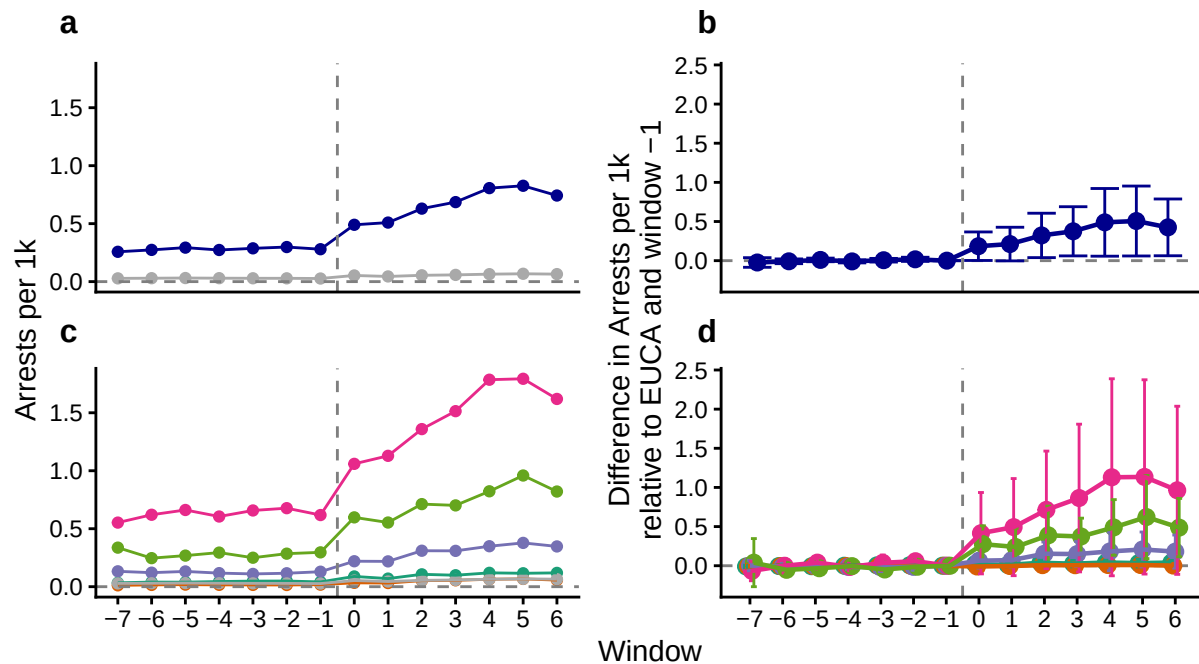


```
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by state, region, window, convicted,
##   threat_level, method, and alt_method.
## i Output is grouped by state, region, window, convicted, threat_level, and
##   method.
## i Use `summarise(groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(state, region, window, convicted, threat_level,
##   method, alt_method))` for per-operation grouping (`?dplyr::dplyr_by`)
##   instead.
```

```
## # A tibble: 70 x 6
##   window     est      se    lower  upper region
##   <dbl>   <dbl>   <dbl>   <dbl>   <dbl> <chr>
## 1     -7 -0.00924 0.00541 -0.0247  0.00618 Africa
## 2     -6 -0.00419 0.00478 -0.0178  0.00942 Africa
## 3     -5 -0.00646 0.00593 -0.0234  0.0104  Africa
## 4     -4  0.000150 0.00615 -0.0174  0.0177  Africa
## 5     -3  0.00475 0.00582 -0.0118  0.0213  Africa
## 6     -2  0.00584 0.00756 -0.0157  0.0274  Africa
## 7     -1  0         0         0         0      Africa
## 8      0  0.0208 0.00709  0.000566 0.0410  Africa
## 9      1  0.0108 0.00865 -0.0139  0.0354  Africa
## 10     2  0.0377 0.0113  0.00555  0.0698  Africa
## # i 60 more rows
## # A tibble: 14 x 6
##   window     est      se    lower  upper region
##   <dbl>   <dbl>   <dbl>   <dbl>   <dbl> <chr>
## 1     -7 -0.0235 0.0238 -0.0849  0.0379 nonEUCA
## 2     -6 -0.00839 0.0118 -0.0387  0.0219 nonEUCA
## 3     -5  0.00947 0.00900 -0.0137  0.0327 nonEUCA
```

```
## 4      -4 -0.00948 0.00751 -0.0289 0.00990 nonEUCA
## 5      -3 0.00560 0.00903 -0.0177 0.0289 nonEUCA
## 6      -2 0.0170 0.0102 -0.00927 0.0433 nonEUCA
## 7      -1 0 0 0 0 nonEUCA
## 8       0 0.184 0.0707 0.00215 0.367 nonEUCA
## 9       1 0.212 0.0833 -0.00302 0.427 nonEUCA
## 10      2 0.322 0.110 0.0388 0.606 nonEUCA
## 11      3 0.375 0.122 0.0609 0.689 nonEUCA
## 12      4 0.489 0.168 0.0570 0.922 nonEUCA
## 13      5 0.506 0.173 0.0597 0.953 nonEUCA
## 14      6 0.425 0.141 0.0625 0.788 nonEUCA
```

```
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```

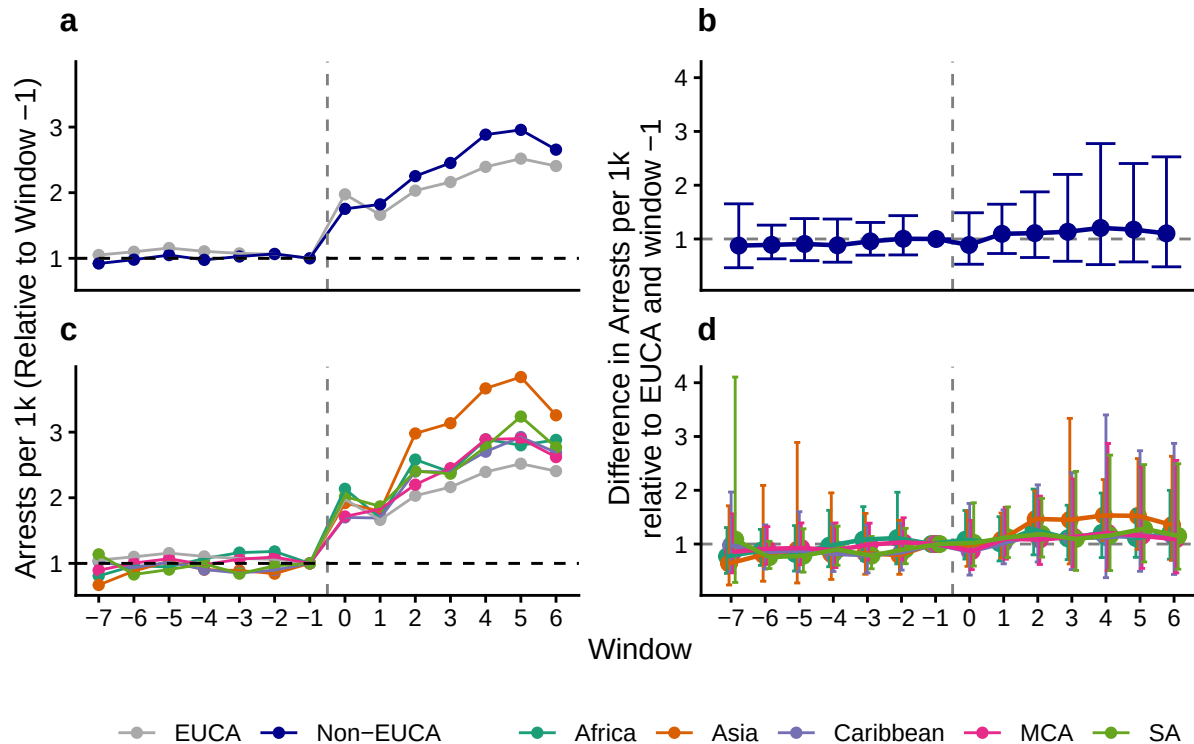


— EUCA — Non-EUCA — Africa — Asia — Caribbean — MCA — SA

```
## # A tibble: 70 x 6
##   window  est    se lower upper region
##   <dbl> <dbl> <dbl> <dbl> <dbl> <chr>
## 1     -7 0.771 0.190 0.454 1.31 Africa
## 2     -6 0.876 0.134 0.603 1.27 Africa
## 3     -5 0.818 0.179 0.498 1.34 Africa
## 4     -4 0.970 0.187 0.578 1.63 Africa
## 5     -3 1.08 0.162 0.691 1.70 Africa
## 6     -2 1.11 0.205 0.630 1.96 Africa
## 7     -1 1 0 1 1 Africa
## 8      0 1.08 0.146 0.721 1.62 Africa
## 9      1 1.02 0.142 0.686 1.51 Africa
## 10     2 1.27 0.167 0.799 2.02 Africa
## # i 60 more rows
```

```
## # A tibble: 14 x 6
##   window  est    se lower upper region
##   <dbl> <dbl> <dbl> <dbl> <dbl> <chr>
## 1     -7 0.877 0.236 0.465  1.65 nonEUCA
## 2     -6 0.891 0.129 0.631  1.26 nonEUCA
## 3     -5 0.907 0.156 0.597  1.38 nonEUCA
## 4     -4 0.883 0.164 0.568  1.37 nonEUCA
## 5     -3 0.956 0.117 0.699  1.31 nonEUCA
## 6     -2 1.00  0.133 0.703  1.43 nonEUCA
## 7     -1 1      0      1      1 nonEUCA
## 8      0 0.888 0.192 0.530  1.49 nonEUCA
## 9      1 1.10  0.152 0.730  1.65 nonEUCA
## 10     2 1.11  0.196 0.655  1.88 nonEUCA
## 11     3 1.13  0.247 0.585  2.20 nonEUCA
## 12     4 1.21  0.311 0.524  2.77 nonEUCA
## 13     5 1.17  0.267 0.574  2.40 nonEUCA
## 14     6 1.10  0.309 0.482  2.53 nonEUCA

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```



```
## Rows: 231336 Columns: 10
## -- Column specification -----
## Delimiter: ","
## chr (4): state, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

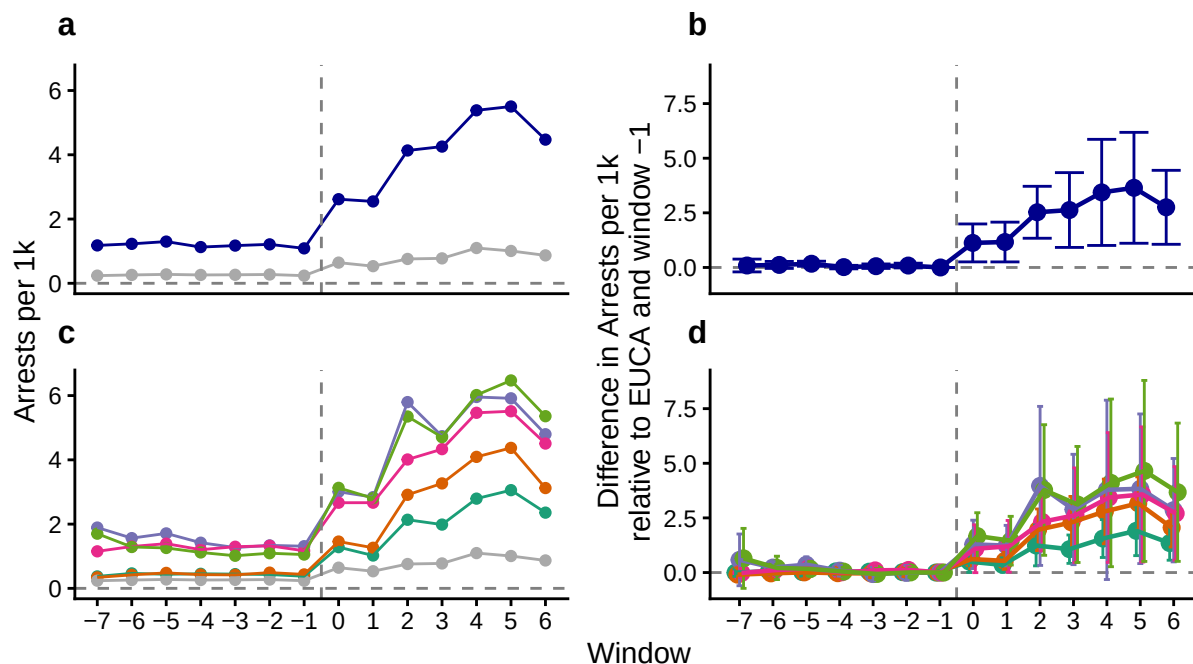
```

## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.

## # A tibble: 70 x 6
##   window     est     se   lower upper region
##   <dbl>   <dbl> <dbl>   <dbl> <dbl> <chr>
## 1     -7 -0.00607 0.0610 -0.177  0.165 Africa
## 2     -6  0.0658  0.0494 -0.0722  0.204 Africa
## 3     -5  0.0216  0.0672 -0.166  0.209 Africa
## 4     -4  0.0535  0.0505 -0.0878  0.195 Africa
## 5     -3  0.0391  0.0387 -0.0692  0.147 Africa
## 6     -2  0.0177  0.0436 -0.104  0.140 Africa
## 7     -1  0      0      0      0      Africa
## 8      0  0.500  0.175  0.00945 0.991 Africa
## 9      1  0.350  0.116  0.0260  0.673 Africa
## 10     2  1.24   0.333  0.307  2.17  Africa
## # i 60 more rows
## # A tibble: 14 x 6
##   window     est     se   lower upper region
##   <dbl>   <dbl> <dbl>   <dbl> <dbl> <chr>
## 1     -7  0.0907 0.111  -0.202  0.384 nonEUCA
## 2     -6  0.117  0.0604 -0.0426  0.277 nonEUCA
## 3     -5  0.171  0.0436  0.0554  0.287 nonEUCA
## 4     -4  0.0173 0.0302 -0.0627  0.0974 nonEUCA
## 5     -3  0.0587 0.0387 -0.0440  0.161 nonEUCA
## 6     -2  0.0881 0.0410 -0.0207  0.197 nonEUCA
## 7     -1  0      0      0      0      nonEUCA
## 8      0  1.12   0.327  0.256  1.99  nonEUCA
## 9      1  1.17   0.343  0.256  2.08  nonEUCA
## 10     2  2.53   0.448  1.34   3.71  nonEUCA
## 11     3  2.63   0.646  0.917  4.34  nonEUCA
## 12     4  3.44   0.916  1.01   5.86  nonEUCA
## 13     5  3.65   0.957  1.11   6.18  nonEUCA
## 14     6  2.75   0.638  1.06   4.45  nonEUCA

## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.

```

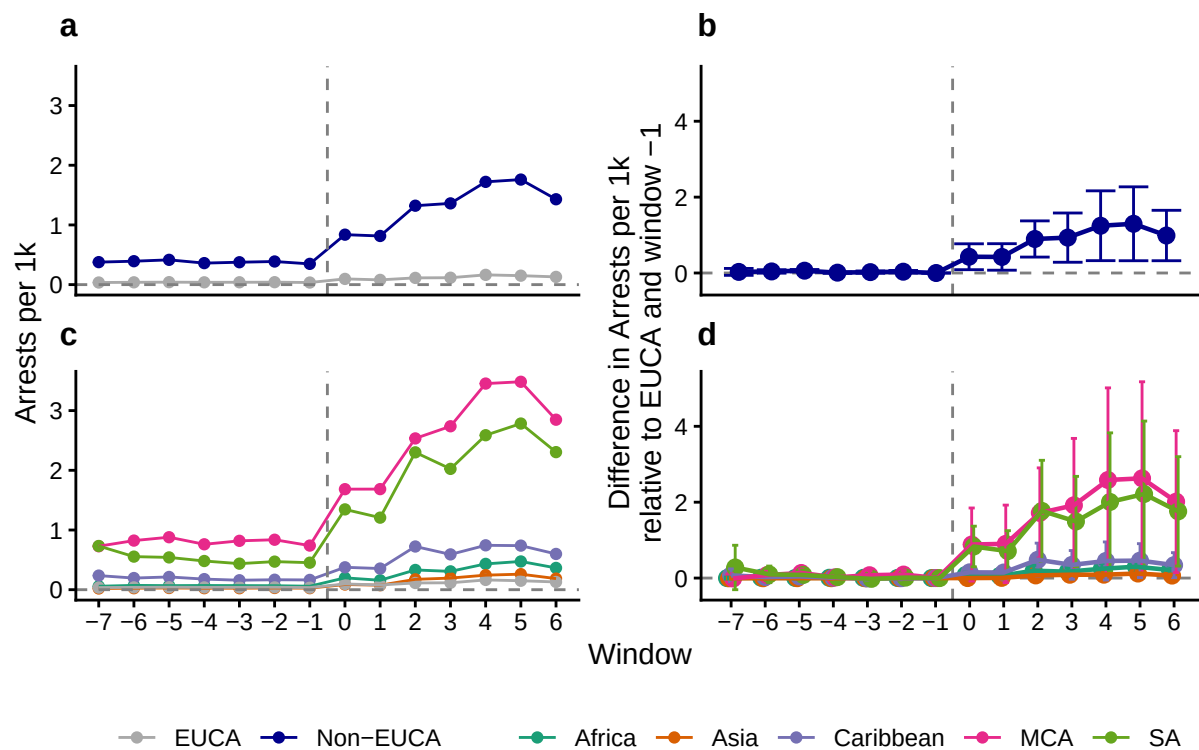
— EUCA — Non-EUCA — Africa — Asia — Caribbean — MCA — SA

```
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by state, region, window, convicted,
##   threat_level, method, and alt_method.
## i Output is grouped by state, region, window, convicted, threat_level, and
##   method.
## i Use `summarise(groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(state, region, window, convicted, threat_level,
##   method, alt_method))` for per-operation grouping (`?dplyr::dplyr_by`)
```

```
## # A tibble: 70 x 6
##   window     est      se   lower upper region
##   <dbl>   <dbl>   <dbl>   <dbl> <dbl> <chr>
## 1     -7 -0.000933 0.00942 -0.0276  0.0257 Africa
## 2     -6  0.0103   0.00752 -0.0110  0.0316 Africa
## 3     -5  0.00357   0.0103  -0.0257  0.0328 Africa
## 4     -4  0.00841   0.00766 -0.0133  0.0301 Africa
## 5     -3  0.00621   0.00594 -0.0106  0.0230 Africa
## 6     -2  0.00294   0.00682 -0.0164  0.0222 Africa
## 7     -1  0         0         0         0 Africa
## 8      0  0.0797   0.0262   0.00545  0.154 Africa
## 9      1  0.0558   0.0181   0.00454  0.107 Africa
## 10     2  0.195    0.0498   0.0536  0.336 Africa
## # i 60 more rows
## # A tibble: 14 x 6
##   window     est      se   lower upper region
##   <dbl>   <dbl>   <dbl>   <dbl> <dbl> <chr>
## 1     -7  0.0292   0.0350  -0.0617  0.120 nonEUCA
## 2     -6  0.0414   0.0174  -0.00370  0.0866 nonEUCA
## 3     -5  0.0615   0.0120   0.0304  0.0926 nonEUCA
```

```
## 4      -4 0.00944 0.00710 -0.00901 0.0279 nonEUCA
## 5      -3 0.0235  0.0117  -0.00701 0.0539 nonEUCA
## 6      -2 0.0344  0.0126   0.00176 0.0670 nonEUCA
## 7      -1 0       0         0       0      nonEUCA
## 8       0 0.429   0.131    0.0875  0.770  nonEUCA
## 9       1 0.423   0.134    0.0738  0.772  nonEUCA
## 10      2 0.897   0.184    0.420   1.37   nonEUCA
## 11      3 0.933   0.250    0.284   1.58   nonEUCA
## 12      4 1.25    0.354    0.325   2.17   nonEUCA
## 13      5 1.30    0.375    0.323   2.27   nonEUCA
## 14      6 0.988   0.256    0.324   1.65   nonEUCA
```

```
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```



```
## # A tibble: 70 x 6
##   window  est    se lower upper region
##   <dbl> <dbl> <dbl> <dbl> <dbl> <chr>
## 1     -7 0.982 0.204 0.559  1.73 Africa
## 2     -6 1.13  0.122 0.803  1.58 Africa
## 3     -5 0.996 0.190 0.589  1.68 Africa
## 4     -4 1.10  0.148 0.728  1.65 Africa
## 5     -3 1.05  0.112 0.773  1.44 Africa
## 6     -2 0.991 0.146 0.662  1.48 Africa
## 7     -1 1      0      1      1      Africa
## 8      0 1.26  0.185 0.751  2.10 Africa
## 9      1 1.21  0.151 0.795  1.84 Africa
## 10     2 1.78  0.218 0.970  3.25 Africa
## # i 60 more rows
```

```
## # A tibble: 14 x 6
##   window  est      se lower upper region
##   <dbl> <dbl> <dbl> <dbl> <dbl> <chr>
## 1     -7 1.08 0.181 0.663 1.76 nonEUCA
## 2     -6 1.03 0.115 0.755 1.40 nonEUCA
## 3     -5 1.02 0.146 0.688 1.52 nonEUCA
## 4     -4 0.946 0.119 0.686 1.31 nonEUCA
## 5     -3 0.968 0.0854 0.768 1.22 nonEUCA
## 6     -2 0.966 0.104 0.729 1.28 nonEUCA
## 7     -1 1      0      1      1 nonEUCA
## 8      0 0.888 0.144 0.603 1.31 nonEUCA
## 9      1 1.04 0.115 0.766 1.42 nonEUCA
## 10     2 1.19 0.166 0.760 1.87 nonEUCA
## 11     3 1.20 0.156 0.785 1.83 nonEUCA
## 12     4 1.07 0.221 0.589 1.95 nonEUCA
## 13     5 1.19 0.168 0.759 1.88 nonEUCA
## 14     6 1.12 0.208 0.642 1.97 nonEUCA
```

```
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```

